*(iii) The functor* $\mathrm{Hom}_A(?, I)$ *on the category of* $A$*-modules is exact.*

**Proof.** The proof of the implications "(i) implies (ii)" and "(iii) implies (i)" is similar to the proof of Theorem 8.1.1. Let us prove that (ii) implies (iii). Let

$$0 \to N \to M \to K \to 0$$

be an exact sequence. Denote the embedding $N \to M$ by $j$. Consider the corresponding sequence

$$0 \to \mathrm{Hom}(K, I) \to \mathrm{Hom}(M, I) \to \mathrm{Hom}(N, I) \to 0.$$

Let $f \in \mathrm{Hom}(N, I)$, and define the module $E := (M \oplus I)/N$, where $N$ is embedded into $M \oplus I$ via $x \mapsto (j(x), -f(x))$. Clearly, we have an inclusion $I \to E$, since the image of $N \oplus I$ in $E$ is naturally identified with $I$. So there is a splitting $E \to I$ of this inclusion, i.e., a map $M \oplus I \to I$, $(m, i) \mapsto g(m) + i$ such that $g(j(x)) - f(x) = 0$. This means that the map $j^* : \mathrm{Hom}(M, I) \to \mathrm{Hom}(N, I)$ is surjective, i.e., the functor $\mathrm{Hom}(?, I)$ is exact, as desired. $\square$

**Definition 8.1.6.** A module satisfying any of the conditions (i)—(iii) of Theorem 8.1.5 is said to be **injective**.

**Example 8.1.7.** Let $A$ be an algebra and $P$ be a left $A$-module. Then $P$ is projective if and only if $P^*$ is an injective right $A$-module.

Theorem 8.1.1(iv) and Theorem 8.1.5(iii) can be used to define a projective and an injective object in any abelian category. Namely, we make the following definition.

**Definition 8.1.8.** A **projective object** in an abelian category $\mathcal{C}$ is an object $P$ such that the functor $\mathrm{Hom}_{\mathcal{C}}(P, ?)$ is exact.

An **injective object** in an abelian category $\mathcal{C}$ is an object $I$ such that the functor $\mathrm{Hom}_{\mathcal{C}}(?, I)$ is exact.

## 8.2. Tor and Ext functors

Let $A$ be a unital ring. As we have mentioned in Example 7.9.6, the functors $M \otimes_A ?$ and $\mathrm{Hom}_A(M, ?)$ (where $M$ is a right, respectively